



JUL 2026

Predmet: Računarska inteligencija

PREDIKCIJA SEKUNDARNE STRUKTURE PROTEINA



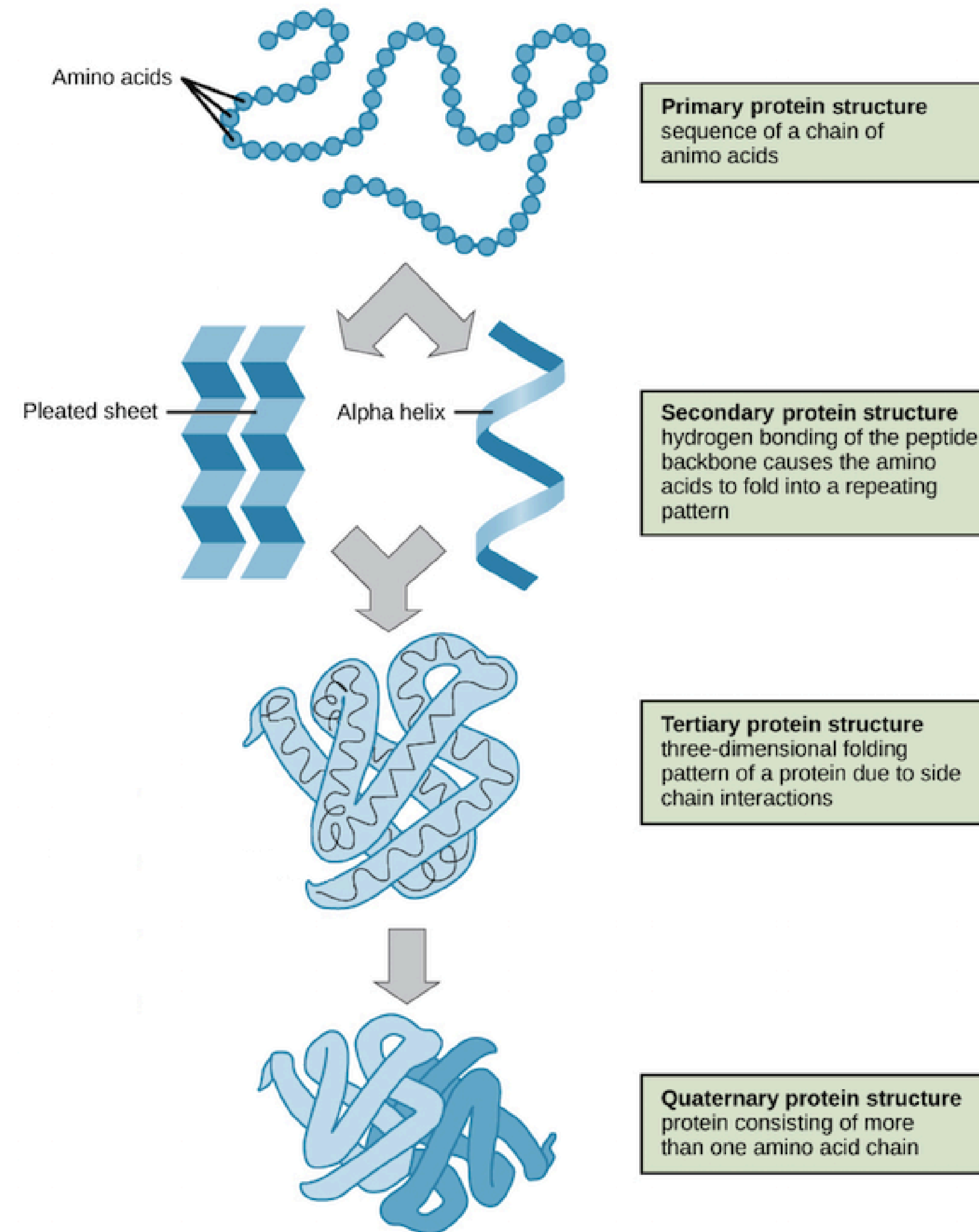
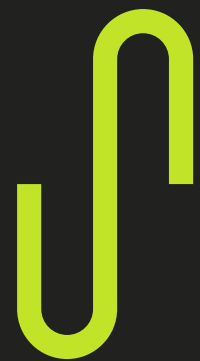
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UVOD

- Proteini - biomolekuli, različite uloge
- Strukture proteina
 - Primarna - niz aminokiselina
 - Sekundarna - α -heliksi, β -ploče
 - Tercijarna (+ kvartenarna)
- Aminokiseline

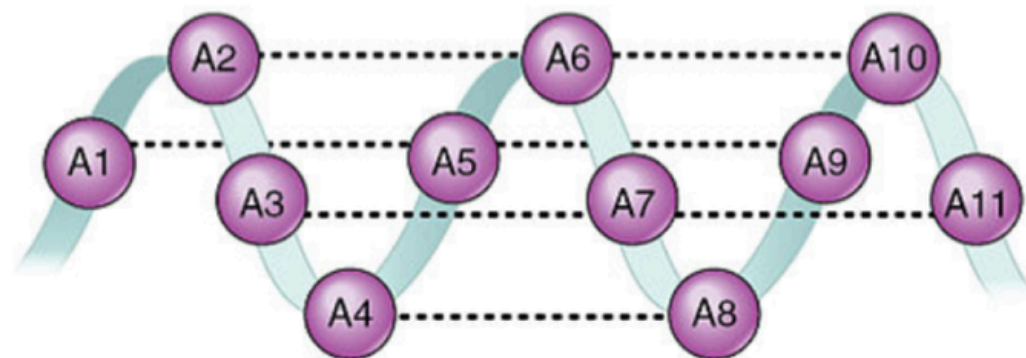




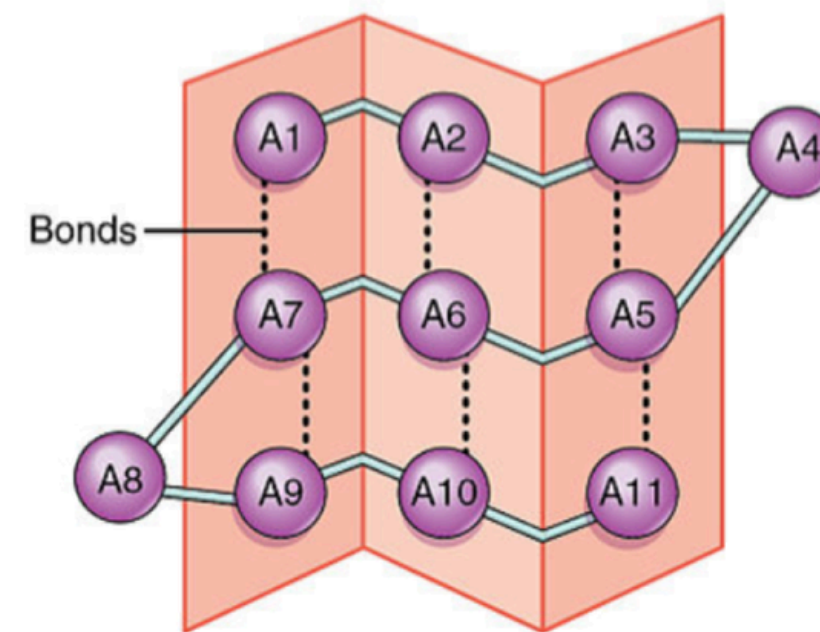
CILJ PROJEKTA



Na osnovu sekvence aminokiselina (primarna struktura) odrediti sekundarnu strukturu (odrediti gde su α -heliksi, β -ploče i petlje)



α - helix



β - pleated sheet



SKUP PODATAKA



Korišćena su 2 skupa podataka:

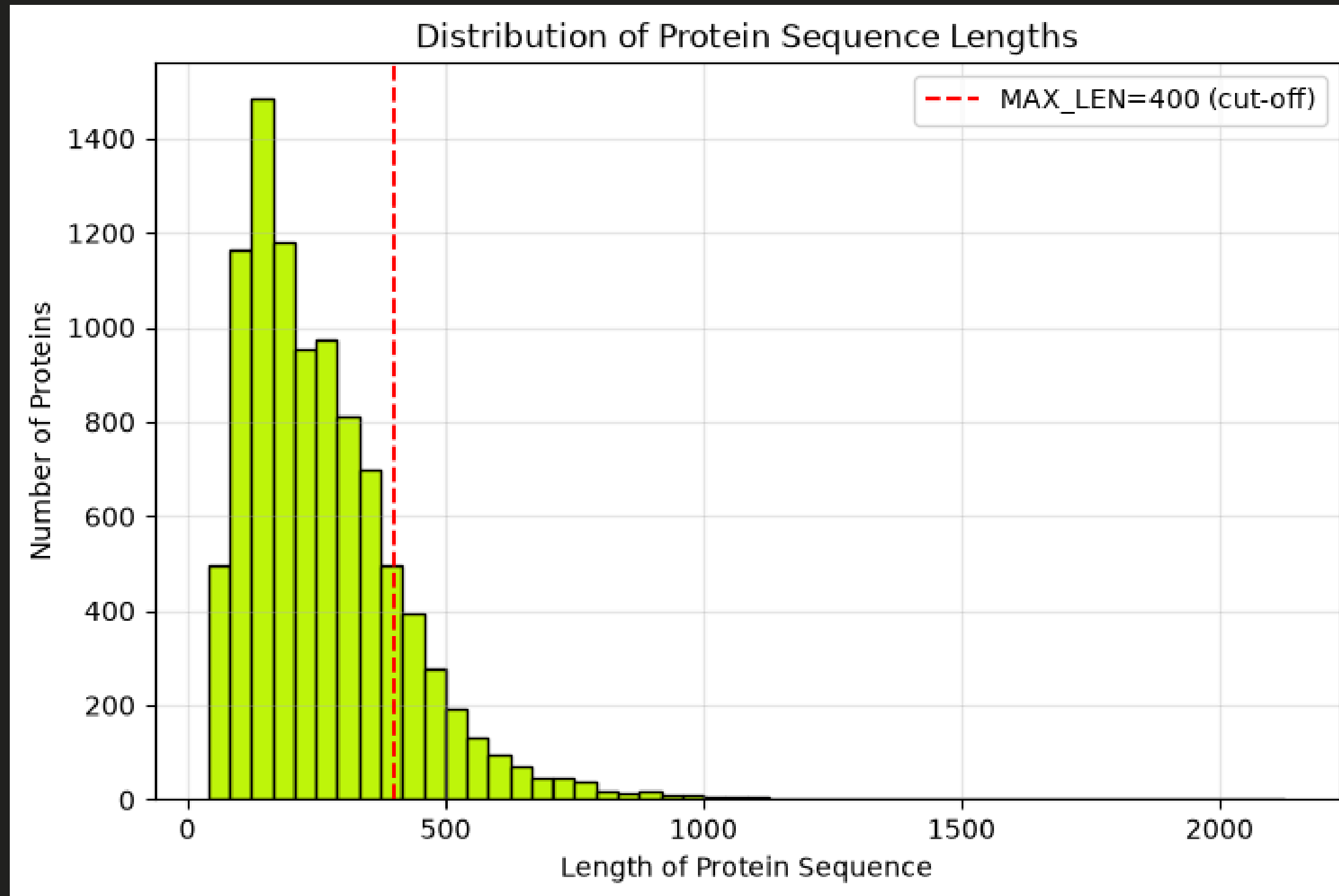
- [CB513](#)
- [Pisces x PDB dataset](#)

Što se tiče samih dataset-ova, glavna obeležja koja su posmatrana su sekvenca aminokiselina (*sequence*) i oznaka sekundarne strukture proteina (*sst3*)

SST-8 Category	Description	SST-3 Category
E	β -strand	E
B	β -bridge	E
H	α -helix	H
G	3-helix	H
I	π -helix	H
C	Loops and irregular elements	C
T	Turn	C
S	Bend	C

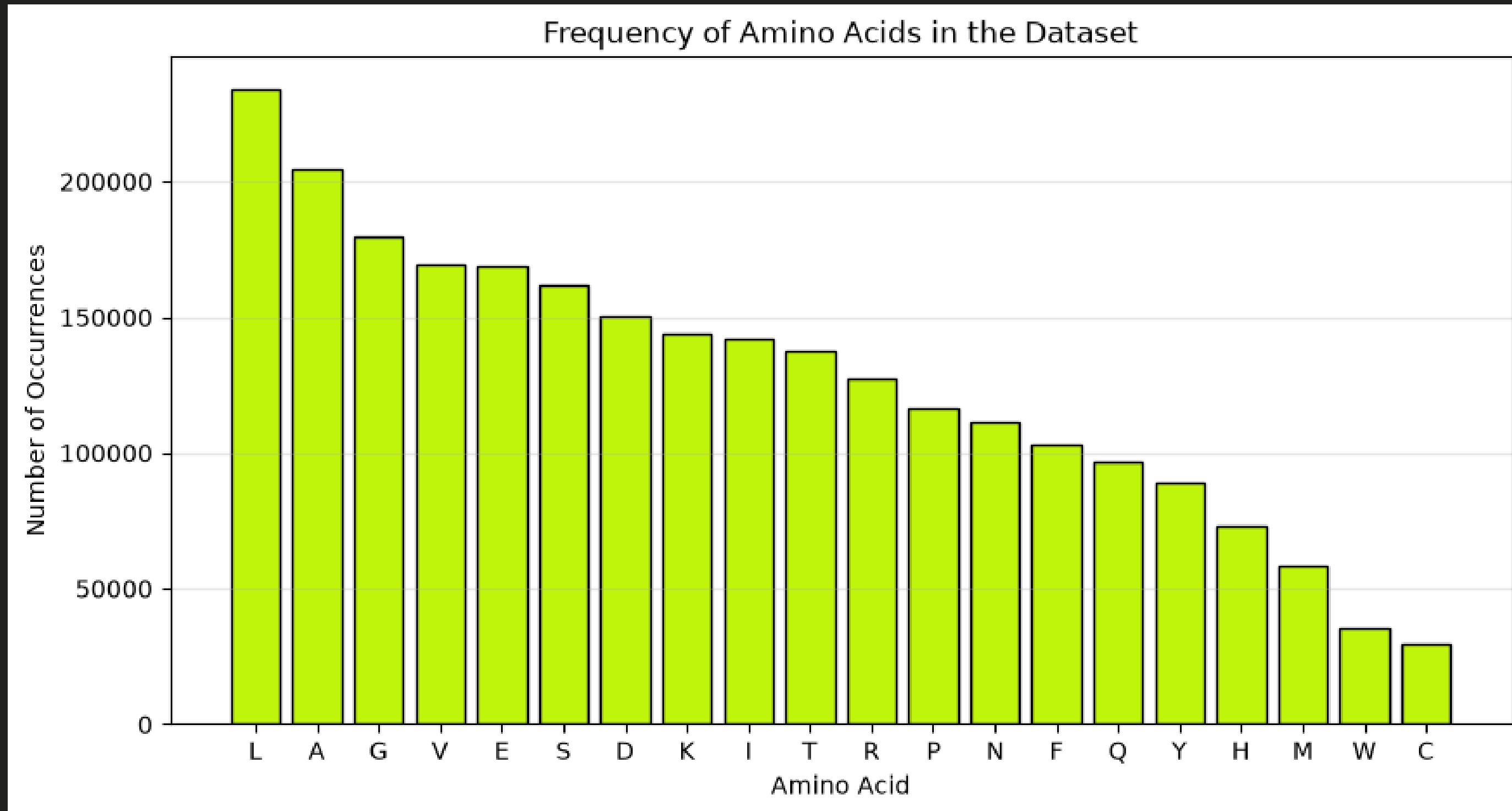


EKSPLORATIVNA ANALIZA DATASETA



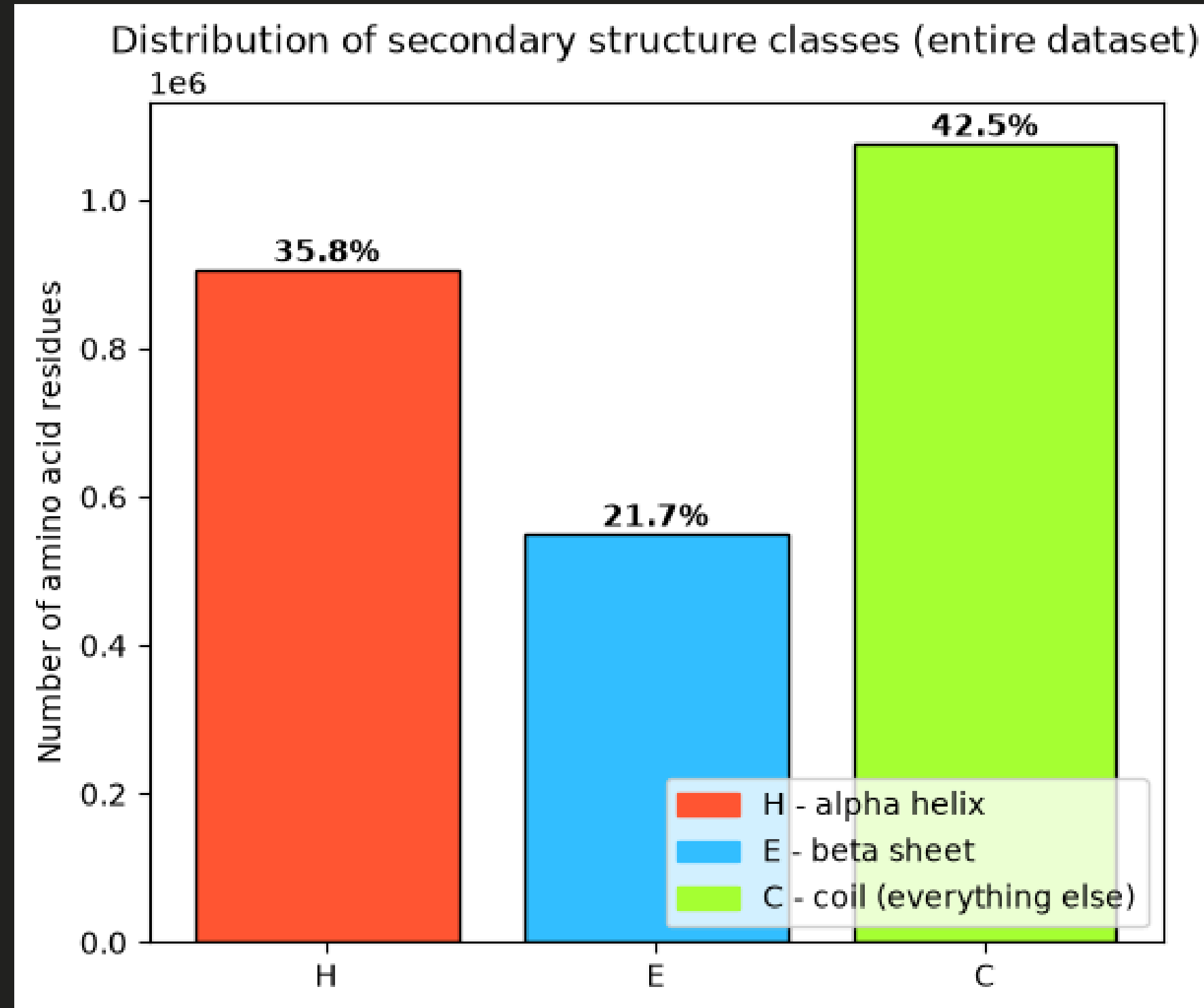


EKSPLORATIVNA ANALIZA DATASETA





EKSPLORATIVNA ANALIZA DATASETA





METODOLOGIJA

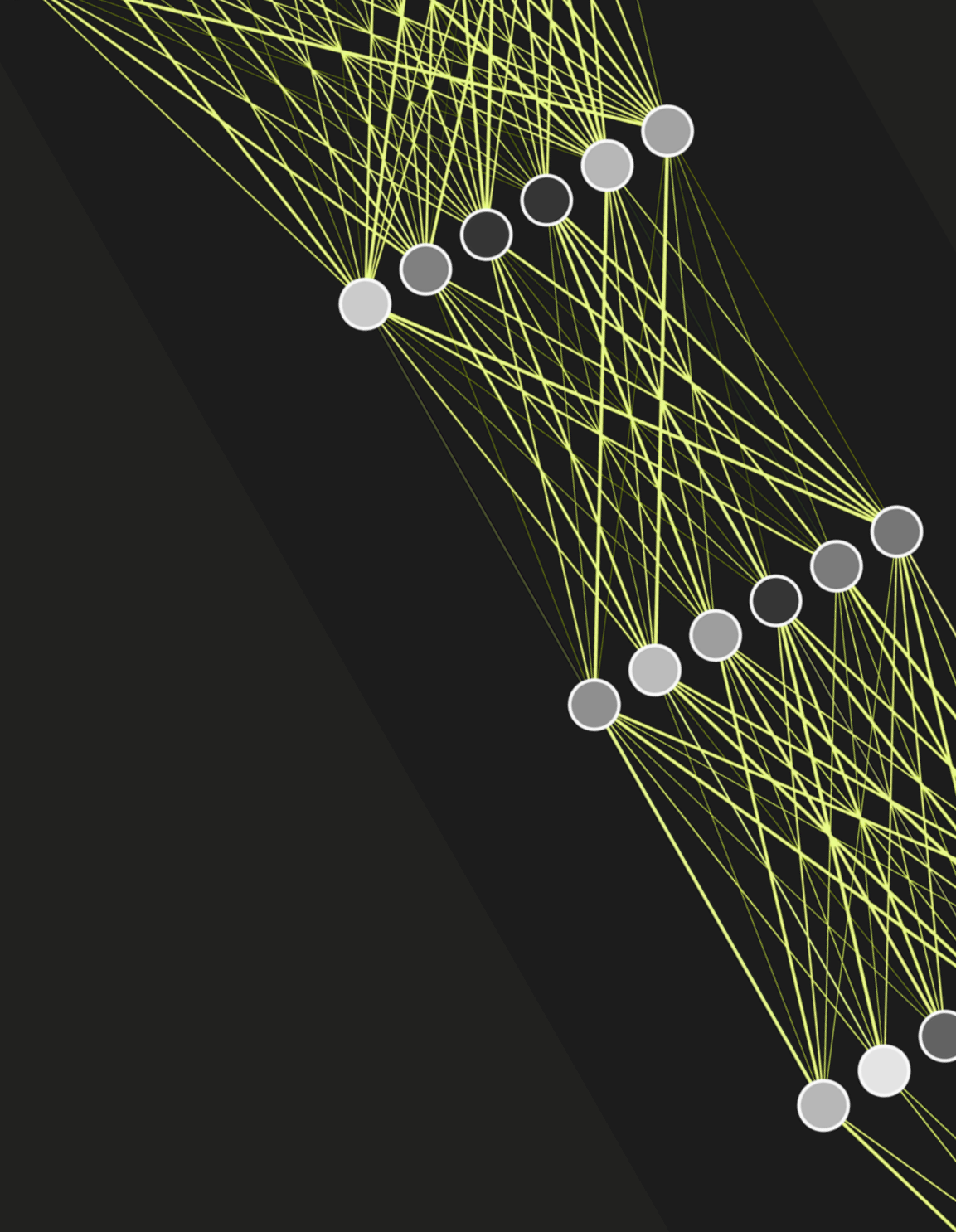
Arhitektura: CNN + BiLSTM

- CNN hvata lokalne motive sekvence
- BiLSTM hvata dugoročni kontekst u oba smera
- Softmax izlaz: H / E / C po ostatku

Optimizacija hiperparametara, poređenje 4 pristupa

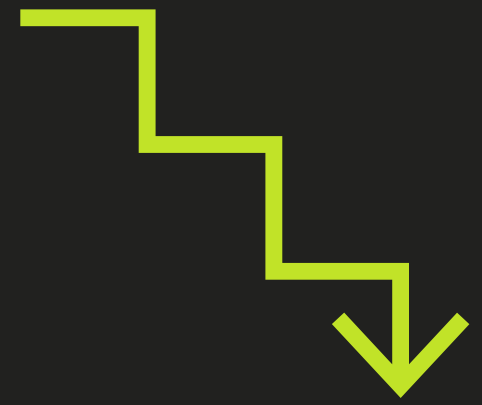
- Random Search (baseline)
- Genetski algoritam (evolucija populacije)
- PSO (roj čestica)
- *Bayesova* optimizacija (Gaussian Process)

Hiperparametri: broj filtera, kernel, LSTM jedinice, learning rate, dropout, batch size





MODEL



- **CNN** sloj hvata lokalne motive (prozor od 5 suseda), "same" padding čuva dužinu sekvence
- **BiLSTM** sloj hvata dugoročni kontekst u oba smera niza (bidirekciono)
- Izlaz po ostatku model predviđa H/E/C za svaki aminokiselinski ostatak posebno (*sequence-to-sequence*, ne klasifikacija cele sekvence)
- Trening - ADAM optimizator, Cross-Entropy loss sa maskiranjem padding pozicija
- **6 hiperparametara** koji se dalje optimizuju: broj filtera, veličina kernela, broj LSTM jedinica, dropout, stopa učenja, veličina batch-a



OPTIMIZACIJE

To ensure long-term sustainability and growth, our business model is designed to generate multiple streams of revenue while delivering value to users.



PREDLOG ZA DALJE



To successfully introduce our product to the market and gain early traction, we've developed a clear go-to-market strategy.





ZAKLJUČAK

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**HVALA NA
PAŽŇNJI :)**

